

RC CAR TEST WORKSHEET

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This worksheet is for the current RC car build:

- front ultrasonic only
- no right ultrasonic
- USB webcam
- LiDAR
- hall sensor
- controller-driven manual mode first

Use one sheet per test session. Change only one tuning value at a time.

1. SESSION INFO

- Test location: _____
- Surface: _____
- Lighting: sun / shade / mixed
- Battery level at start: _____
- Notes: _____

2. CURRENT VARIABLE REFERENCES

Main tuning constants in [config.py](/Users/ramsabavat/Desktop/RC%20CAR%20CODE/CODE/CONTROLLER/CURRENT/rc_car_app/config.py):

- `ACCEL\_RATE`
- `BRAKE\_RATE`
- `COASTING\_RATE`
- `AEB\_BRAKE\_RATE`
- `FRONT\_FAILSAFE\_STOP\_CM`
- `AUTONOMOUS\_CRUISE\_PWM`
- `AUTONOMOUS\_TURN\_PWM`
- `AUTONOMOUS\_CAUTIOUS\_PWM`
- `MAX\_TARGET\_HEADING\_DEG`
- `LOW\_CAMERA\_CONFIDENCE`
- `HIGH\_CAMERA\_CONFIDENCE`
- `FORWARD\_OBSTACLE\_STOP\_DISTANCE\_M`
- `PARTIAL\_BLOCKAGE\_MIN\_CLEARANCE\_M`
- `DRIVEWAY\_CUT\_MIN\_FORWARD\_CLEARANCE\_M`
- `DRIVEWAY\_CUT\_DETECTION\_SECONDS`
- `LIDAR\_FORWARD\_ANGLE\_MIN\_DEG`
- `LIDAR\_FORWARD\_ANGLE\_MAX\_DEG`
- `LIDAR\_HEADING\_WINDOW\_DEG`
- `WHEEL\_DIAMETER\_CM`
- `PULSES\_PER\_REVOLUTION`
- `SPEED\_SMOOTHING\_ALPHA`
- `KP`
- `KI`
- `KD`

Useful live state fields in [runtime.py](/Users/ramsabavat/Desktop/RC%20CAR%20CODE/CODE/CONTROLLER/CURRENT/rc_car_app/runtime.py) and [config.py](/Users/ramsabavat/Desktop/RC%20CAR%20CODE/CODE/CONTROLLER/CURRENT/rc_car_app/config.py):

- `state["steer"]`
- `state["throttle"]`
- `state["brake"]`
- `state["gear\_mode"]`
- `state["current\_motor\_pwm"]`
- `state["target\_heading\_deg"]`
- `state["stop\_reason"]`
- `state["lidar\_best\_heading\_deg"]`
- `state["lidar\_heading\_confidence"]`
- `state["lidar\_forward\_clearance\_m"]`
- `state["camera\_steering\_bias"]`

- `state["camera\_confidence"]`
- `state["camera\_left\_edge\_found"]`
- `state["camera\_right\_edge\_found"]`
- `state["camera\_corridor\_width\_px"]`
- `state["front\_ultrasonic\_distance\_cm"]`
- `metrics.current\_raw\_mph`
- `metrics.smoothed\_speed\_mph`
- `metrics.pid\_output`
- `metrics.last\_confident\_heading\_deg`

3. BASE CAR VALIDATION

3.1 STEERING CENTER

- Test: Drive straight 2 to 3 meters with `state["steer"] = 0`.
- Result: drifts left / drifts right / tracks straight
- Notes: _____
- Action:
 - if drift left, adjust steering center slightly right
 - if drift right, adjust steering center slightly left

Related variables:

- `state["steer"]`
- steering servo center calibration in the steering hardware path
- `MAX\_TARGET\_HEADING\_DEG` later for autonomy range, not center trim

3.2 STEERING LIMITS

- Test: Full-left and full-right with wheels off ground, then slow circles on ground.
- Result: binds / no bind / too aggressive / too weak
- Notes: _____

Related variables:

- `MAX\_TARGET\_HEADING\_DEG`
- any servo endpoint limit you set in steering calibration

3.3 FORWARD / REVERSE

- Test: Verify `D` goes forward and `R` goes backward.
- Result: correct / inverted
- Notes: _____

Related variables:

- `state["gear\_mode"]`
- `state["current\_motor\_pwm"]`
- motor direction mapping in
[runtime.py](/Users/ramsabavat/Desktop/RC%20CAR%20CODE/CODE/CONTROLLER/CURRENT/rc_car_app/runtime.py)

3.4 BRAKING

- Test: Slow roll, then brake.
- Result: too weak / good / too harsh
- Notes: _____

Related variables:

- `BRAKE\_RATE`
- `AEB\_BRAKE\_RATE`
- `state["brake"]`

3.5 GEAR CHANGES

- Test: Shift through `P`, `R`, `N`, `D`.

- Result: correct / incorrect
- Notes: _____

Related variables:

- ``GEARS``
- ``state["gear_mode"]``

4. MANUAL-DRIVE TUNING

4.1 ACCELERATION RAMP

- Test: From stop in ``D``, apply throttle gently.
- Result: too jerky / acceptable / too weak
- Notes: _____

Change:

- lower ``ACCEL_RATE`` if jerky
- raise ``ACCEL_RATE`` if too weak

4.2 COASTING

- Test: Release throttle while moving slowly.
- Result: slows too fast / good / rolls too long
- Notes: _____

Change:

- lower ``COASTING_RATE`` if slowing too fast
- raise ``COASTING_RATE`` if it rolls too long

4.3 REVERSE FEEL

- Test: Back up slowly in ``R``.
- Result: manageable / too aggressive / too weak
- Notes: _____

Related variables:

- ``state["gear_mode"]``
- ``state["current_motor_pwm"]``
- reverse logic in `[runtime.py](/Users/ramsabavat/Desktop/RC%20CAR%20CODE/CODE/CONTROLLER/CURRENT/rc_car_app/runtime.py)`

5. SENSOR VALIDATION

5.1 HALL SENSOR

- Test: Move car at known slow speed and compare reported speed trend.
- Result: works / noisy / wrong scale / no response
- Notes: _____

Change:

- ``PULSES_PER_REVOLUTION``
- ``WHEEL_DIAMETER_CM``
- ``SPEED_SMOOTHING_ALPHA``

Watch:

- ``metrics.current_raw_mph``
- ``metrics.smoothed_speed_mph``

5.2 FRONT ULTRASONIC

- Test: Place object at 20 cm, 40 cm, 60 cm.

- Result: believable / noisy / wrong / intermittent
- Notes: _____

Change:

- `FRONT\_FAILSAFE\_STOP\_CM`

Watch:

- `state["front\_ultrasonic\_distance\_cm"]`
- `state["stop\_reason"]`

5.3 LIDAR

- Test: Manual drive straight with open path and static edges.
- Result: stable / noisy / disconnects
- Notes: _____

Change:

- `LIDAR\_FORWARD\_ANGLE\_MIN\_DEG`
- `LIDAR\_FORWARD\_ANGLE\_MAX\_DEG`
- `LIDAR\_HEADING\_WINDOW\_DEG`
- `FORWARD\_OBSTACLE\_STOP\_DISTANCE\_M`
- `PARTIAL\_BLOCKAGE\_MIN\_CLEARANCE\_M`

Watch:

- `state["lidar\_best\_heading\_deg"]`
- `state["lidar\_heading\_confidence"]`
- `state["lidar\_forward\_clearance\_m"]`
- `state["direction\_arrow"]`

5.4 WEBCAM

- Test: Manual drive in sun and shade.
- Result: stable / loses path in shade / overreacts / confidence too low
- Notes: _____

Change:

- `LOW\_CAMERA\_CONFIDENCE`
- `HIGH\_CAMERA\_CONFIDENCE`
- `CAMERA\_STEER\_GAIN`
- `CAMERA\_TURN\_BLEND`

Watch:

- `state["camera\_steering\_bias"]`
- `state["camera\_confidence"]`
- `state["camera\_left\_edge\_found"]`
- `state["camera\_right\_edge\_found"]`
- `state["camera\_corridor\_width\_px"]`

6. FIRST AUTONOMY CHECKS

6.1 STRAIGHT PATH TEST

- Test: Closed course, low speed, straight path.
- Result: stays centered / drifts / oscillates / stops too often
- Notes: _____

Change:

- `MAX\_TARGET\_HEADING\_DEG`
- `AUTONOMOUS\_CAUTIOUS\_PWM`

- `AUTONOMOUS\_CRUISE\_PWM`
- `LOW\_CAMERA\_CONFIDENCE`

6.2 TURN TEST

- Test: Closed course, gentle and sharper turn.
- Result: makes turn / understeers / oversteers / oscillates
- Notes: _____

Change:

- `MAX\_TARGET\_HEADING\_DEG`
- `AUTONOMOUS\_TURN\_PWM`
- `CAMERA\_TURN\_BLEND`

6.3 OBSTACLE STOP TEST

- Test: Approach a box or cone slowly.
- Result: stops too early / good / stops too late
- Notes: _____

Change:

- `FRONT\_FAILSAFE\_STOP\_CM`
- `FORWARD\_OBSTACLE\_STOP\_DISTANCE\_M`
- `PARTIAL\_BLOCKAGE\_MIN\_CLEARANCE\_M`

Watch:

- `state["stop\_reason"]`
- `state["front\_ultrasonic\_distance\_cm"]`
- `state["lidar\_forward\_clearance\_m"]`

7. QUESTIONS TO ANSWER AFTER EACH SESSION

- Does the car track straight with steer = 0? _____
- Is steering too aggressive or too weak? _____
- Does LiDAR heading feel noisy while moving? _____
- Does the webcam keep seeing the sidewalk in shade? _____
- Does the car stop too early or too late? _____
- Does it oscillate left/right? _____

8. CHANGE LOG

- Change 1: _____
- Old value: _____
- New value: _____
- Better / worse / no change

- Change 2: _____
- Old value: _____
- New value: _____
- Better / worse / no change

- Change 3: _____
- Old value: _____
- New value: _____
- Better / worse / no change

9. STOP CONDITIONS

Stop testing immediately if:

- steering becomes unpredictable
- servo jitter changes the path unexpectedly
- LiDAR disconnects repeatedly

- braking is not reliable
- the car leaves the intended test path